

SCIENTIFIC REPORT · STRUCTURAL HEALING · BERTONE ET AL. 2026

Structural healing on quantitative ultrasound.

eCTM ligaments rebuilt tissue architecture over 12 weeks; standard-of-care ligaments deteriorated.

Background

OBJECTIVE

Return to sport depends on structural healing, not just symptom relief. Yet standard B-mode ultrasound reads out qualitatively and offers no reliable early signal of tissue repair. We asked whether a quantitative texture-based Tissue Remodeling Score (TRS) could detect objective architectural recovery after a single injection of cryopreserved umbilical allograft (equicenta® CTM) versus clinician-selected standard of care.

Methods

IMAGING & SCORING

B-mode cine loops (Butterfly iQ3) at Weeks 0, 2, 4 and 12. Six quantitative texture features were combined into a composite TRS, normalized so healthy tissue scores 50 at baseline. Thresholds: Responding (55 and above) · Equivocal (45–55) · Not Responding (<45). Segmentation was blinded to treatment.

TWO-LAYER MODEL OF HEALING

TRS decomposes into MS Core (deep macrostructural scaffold: central dark area, dark intensity) and Quality Layer (fiber alignment, fiber echo intensity). Responder ligaments showed early MS Core recovery by Week 4, followed by late Quality Layer remodeling — deep architecture reorganizing before surface tissue matures.

SAMPLE & VALIDATION

46 QC-passed injuries: 22 eCTM vs. 11 standard-of-care ligaments plus 13 exploratory tendons. Segmentation reliability across 25 ROIs / 8 cases: Dice 0.931. Tendon TRS correlated with limb lameness at Weeks 2–4 ($r=-0.74$, $p<0.02$); ligament TRS trajectory converged with 26-week performance (Cohen's $d=0.96$).

Interpretation

TISSUE-LEVEL EVIDENCE OF HEALING

eCTM ligaments built responding TRS scores (at or above 55) across Weeks 2–12; standard-of-care ligaments drifted below Not-Responding (<45). The healing signature was sequential — MS Core first, Quality Layer following — consistent with a biodegradable allograft releasing reparative proteins over time. Tendon TRS tracked lameness at 2–4 wk ($r=-0.74$), and the ligament TRS trajectory converged with 26-week performance (Cohen's $d=0.96$): structural healing preceded and predicted return to work.

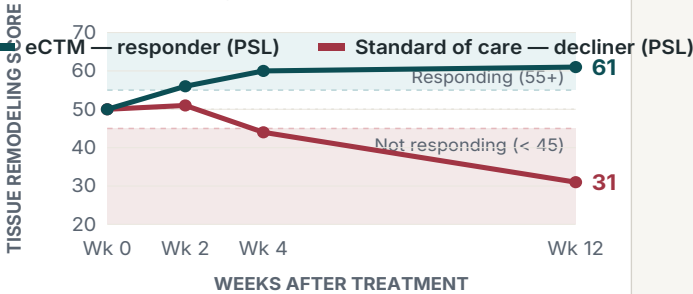
Results

LIGAMENTS REACHING THE RESPONDING TRS THRESHOLD BY WEEK 12

36% : 0%

8 / 22 eCTM vs. 0 / 11 standard of care
 $p=0.013$ · Cohen's $d=0.98$

FIG 1 · Structural healing trajectory (representative cases)



TRS over 12 weeks in matched proximal suspensory ligament (PSL) lesions. The eCTM case rose from baseline (50) into the Responding zone by Week 2 and reached 61; the standard-of-care case dropped below the Not-Responding threshold by Week 4 and reached 31. Illustrative trajectories from Supplementary Figure S1.

Healing you can measure, not just observe.

A blinded quantitative-ultrasound biomarker showed eCTM ligaments rebuilding tissue architecture over 12 weeks — a structural signature standard of care did not produce.

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